Smart IoT Device Troubleshooting AI Agent

Problem Statement Document

1. Project Domain

Internet of Things (IoT) / Artificial Intelligence / Smart Devices

2. Problem Statement

Smart IoT devices are increasingly used in homes, offices, and other environments, but users often face problems such as connectivity failures, Wi-Fi or Bluetooth issues, incorrect configuration, compatibility problems, firmware errors, sensor failures, and unexpected device malfunctions.

Many users do not have the technical knowledge required to identify the root cause of these problems. Troubleshooting information is often scattered across device manuals, technical documentation, support pages, and troubleshooting guides. Users may therefore try unsuitable solutions, repeat basic steps, or contact manufacturer support before simpler checks have been completed.

A conversational AI agent is needed to understand symptoms in natural language, ask relevant questions when important information is missing, retrieve trusted device-specific information, and guide users through safe troubleshooting steps from simple checks to advanced solutions.

3. Key Challenges

- Difficulty identifying the cause of IoT device connectivity and configuration problems.
- Wi-Fi and Bluetooth connection failures or unstable connections.
- Device compatibility and configuration issues.
- Firmware-related errors and outdated software.
- Sensor problems and unexpected device behavior.
- Troubleshooting information spread across multiple technical sources.
- Users may lack the technical knowledge to safely perform advanced troubleshooting.

4. Proposed AI-Based Solution

The proposed solution is a **Smart IoT Device Troubleshooting AI Agent** that works as a conversational assistant. It understands device symptoms described in natural language, asks relevant follow-up questions, retrieves trusted information using Retrieval-Augmented Generation (RAG), and provides clear step-by-step troubleshooting guidance.

The troubleshooting process starts with simple and safe checks, such as power, connectivity, and configuration verification. Only when appropriate does the agent escalate to advanced troubleshooting. When a problem requires professional assistance or manufacturer support, the agent clearly communicates that escalation.

5. Objectives

- Understand IoT device symptoms through natural language conversation.
- Identify problems involving connectivity, Wi-Fi, Bluetooth, configuration, compatibility, firmware, sensors, or device malfunction.

- Ask relevant questions when important diagnostic information is missing.
- Use RAG to retrieve information from trusted IoT manuals and technical documentation.
- Provide clear, safe, step-by-step troubleshooting instructions.
- Start with simple checks and escalate to advanced solutions only when appropriate.
- Track troubleshooting progress and suggest the next appropriate step.
- Clearly identify situations that require manufacturer or professional support.

6. RAG-Based Knowledge Retrieval

The agent uses Retrieval-Augmented Generation (RAG) to retrieve relevant information before generating troubleshooting guidance. Potential knowledge sources include trusted IoT device manuals, technical documentation, manufacturer troubleshooting guides, and device specifications.

RAG helps the agent provide device-relevant responses instead of relying only on general conversational knowledge. Retrieved information can guide configuration checks, compatibility verification, firmware troubleshooting, sensor diagnostics, and other supported procedures.

7. Expected Outcome

The expected outcome is a user-friendly conversational troubleshooting system that helps users diagnose common IoT device problems efficiently and safely. The agent should reduce unnecessary troubleshooting effort by asking relevant questions, retrieving appropriate technical information, guiding users through ordered troubleshooting steps, confirming progress, and escalating to manufacturer or professional support when required.

8. Example Interaction

User: My smart device is not connecting to Wi-Fi.

Agent: Ask for the device model, Wi-Fi status, and whether the device was previously connected.

Agent: Retrieve applicable troubleshooting information from trusted documentation.

Agent: Guide the user through simple checks such as power, Wi-Fi availability, distance, and configuration.

Agent: Confirm whether the issue is resolved and provide the next step if it is not.

9. Safety and Escalation

The agent should prioritize safe troubleshooting. It should begin with simple, low-risk checks and avoid instructing users to perform unsafe electrical, hardware, or manufacturer-restricted procedures. If the available information indicates a hardware failure, safety concern, unsupported modification, or a problem requiring specialized service, the agent should clearly recommend contacting the device manufacturer or a qualified professional.

10. Project Scope

The project focuses on conversational diagnosis and troubleshooting guidance for IoT devices. It is intended to support users with common connectivity, configuration, compatibility, firmware, sensor, and device-operation issues using trusted technical information and a structured troubleshooting workflow.

Project: Smart IoT Device Troubleshooting AI Agent
Domain: Internet of Things (IoT) / Artificial Intelligence
Core Approach: Conversational AI + RAG + Safe Step-by-Step Troubleshooting